

Praveen Korukonda

+91 6302-690832 | praveen.korukonda03@gmail.com | Visakhapatnam, India

Education

Indian Institute of Technology Roorkee , Uttarakhand, India <i>Bachelor of Technology, Mechanical Engineering</i>	Oct 2021 – May 2025 GPA: 7.87/10
Sri Viswa Junior College , Andhra Pradesh, India <i>Higher Secondary School</i>	June 2019 – March 2021 Score: 965/1000
MP & EV English Medium School , Andhra Pradesh, India <i>Secondary School</i>	April 2018 – April 2019 Score: 478/500

Research Interests

Soft and Bio-inspired Robotics, Computer Vision for Robotic Perception, Sensor-driven Machine Learning
Robot Learning and Control, Soft Actuation

Research Experience

Bachelor Thesis | Estimation of Remaining Useful Life of Bearings Using ML Sep 2024 – May 2025
Supervisor: [Prof. S.P. Harsha](#)

- Built a predictive-maintenance framework estimating Remaining Useful Life of high-speed bearings
- Collected vibration signals from an experimental test rig and denoised them via wavelet thresholding
- Extracted 34 multi-domain features and filtered them via S-weighted and Pearson correlation scoring
- Designed a self-attention-augmented BiLSTM to model temporal degradation patterns in vibration data
- Tuned hyperparameters via Bayesian optimisation, selecting hidden units, dropout, and learning rate
- Achieved R^2 of 0.825 on combined data and 0.781 on the held-out bearing under leave-one-out testing

Research Internship | Simulation of Human Induced Load Patterns Using ML July 2024 – Aug 2024
Supervisor: [Prof. Anil Kumar](#)

- Developed a GAN-based model generating human-induced load patterns at 98% similarity to real data
- Processed 658+ experimental samples across walking, jumping, and bouncing loads at 1.5 to 3.5 Hz
- Built a Forecasting+LSTM hybrid model and studied ARIMA and Prophet for time-series load synthesis
- Validated synthetic loads via PSD replication and statistical equivalence in structural response

Lab Based Study | Design and Analysis of Inflatable Decoy Jan 2024 – April 2024
Supervisor: [Prof. S.H. Upadhyay](#)

- Designed a 1/12-scaled inflatable decoy of a missile launcher using a PVC-coated polyester fabric
- Compared box, baffle, and pipe geometries via ABAQUS FEA, choosing pipe structure for bulge control
- Simulated inflation–deflation cycles in ABAQUS using the fabric material model and fluid cavities
- Built physical prototype using 3D-printed joints and tested radar and thermal-IR signature samples

Projects

Design and Analysis of Programmable Soft Robot | *Robotics Hackathon, IIT Roorkee*

- Designed a soft biomimetic crawler inspired by worm and caterpillar peristaltic-undulatory motion
- Modelled the body geometry in SolidWorks with focus on flexibility, compliance, and miniaturisation
- Ran ANSYS FEA to simulate deformation under actuation and validate the soft body's crawling gait
- Engineered programmable magnetisation patterns inside the body for external-field-driven crawling

CAE Design and Structural Analysis of Roll Cage | Supervisor: [Indra Vir Singh](#)

- Reviewed roll cage design standards and prior reports to anchor the project's safety and load brief
- Built and iteratively optimised welded CAD models of the roll cage to maximise structural strength
- Ran ANSYS impact simulations to identify deformation modes and address structural safety concerns

Code Development of 1-D Fanno Flow | Supervisor: [Prof. Ankit Bansal](#)

- Built a MATLAB tool simulating 1-D Fanno flow with shock waves and frictional gas-dynamics effects
- Implemented governing equations, validated against analytical results, and tuned numerical stability

Vibration Control of Supported Beams | Supervisor: [Prof. M.M. Joglekar](#)

- Modelled a supported beam with and without an attached spring-mass-damper subsystem at the midspan
- Ran MATLAB simulations to compare dynamic responses and quantify the SMD's vibration suppression

Image Denoiser | Vision and Language Group (VLG), IIT Roorkee

[github/image-denoiser](#)

- Designed an autoencoder-based image denoiser for low-light photos, reaching a PSNR of 28.256 dB
- Applied RIDNet to improve visibility and detail retention, raising PSNR to 30.579 dB on the same set

Stock Sentiment Analysis Using ML | [Finance Club](#), IIT Roorkee

[github/stock-sentiment](#)

- Built an NLP-driven sentiment model to predict stock-price movements from financial news headlines
- Back-tested the trading strategy on historical data, yielding 17.12% returns and a 1.34 Sharpe ratio

Byteshell | Association for Computing Machinery (ACM), IIT Roorkee

[praveenk.dev/byte-shell](#)

- Built a Unix command line shell from scratch in C that used to execute built-in Linux commands
- Implemented core features like directory navigation and process handling using low-level unix system calls

Professional Experience

Software Engineering Intern | [Planto AI](#)

Sept 2025 – Dec 2025

- Built an incident management system using microservices architecture with Go and PostgreSQL database
- Designed 15+ REST APIs with unit tests across 4 services to manage end-to-end incident lifecycles
- Reduced Mean Time to Acknowledge (MTTA) by 27% through automated, real-time Slack alerts
- Set up background cron jobs for automated incident assignment and send frequent reminders
- Built an invoice generation service in Go to create dynamic PDF invoices, achieving 55 ms average latency

Software Engineering Intern | [CodingPro](#)

June 2025 – Aug 2025

- Built an end-to-end secure job portal application using Spring Boot, Thymeleaf and MySQL database
- Containerized the application using Docker and deployed it to AWS Elastic Beanstalk with RDS for storage
- Implemented role-based authentication with Spring Security to secure both candidate and recruiter profiles
- Developed e2e recruitment workflows, profile management, job application tracking, and resume uploads

Honors & Achievements

- Secured an All-India-Rank of 6026 in JEE Main 2021 amongst 1.5 Million candidates
- Secured an All-India-Rank of 7514 in JEE Advanced 2021 amongst 250,000 candidates
- Awarded Merit-cum-Means (MCM) Scholarship for four consecutive years at IIT Roorkee
- Led a 60-student contingent at Inter Hostel Tech Fest at IIT Roorkee, securing 4 silver medals

Relevant Courses

Kinematics of Machines (MIN-201), Dynamics of Machines (MIN-301), Machine Design (MIN-302)
Statistical Machine Learning (MIL-505), Artificial Intelligence (MIN-516), Machine Drawing (MIN-204)
Programming & Data Structures (MIN-103), Engineering Analysis & Design (MIN-291)

Skills

Languages: Python, C/C++, Java, Go

Tools & Technologies: SolidWorks, Ansys, Abaqus, Matlab, AutoCAD, TensorFlow, PyTorch, Git, Pandas